Fraud Detection in Credit Card Management

Powered By

IBM watsonx.data and IBM BOB

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"Transforming Risk into Resilience"

The Challenge

- Rising cases of **credit card fraud**
- Traditional rule-based systems → too many false positives
- Customer frustration and operational inefficiency

Solution

- The bank turned to **IBM watsonx.data** and **IBM BOB** to modernize its fraud detection strategy:

  - **watsonx.data**: Provided a **lakehouse architecture** that unified structured transaction data, unstructured customer interactions, and external feeds (e.g., dark web monitoring). This allowed analysts to query massive datasets in real time without moving data across silos.

  - **IBM BOB**: Acted as the **command center** for fraud operations. It integrated alerts, automated workflows, and AI-driven recommendations into a single dashboard, enabling fraud teams to respond faster and smarter.

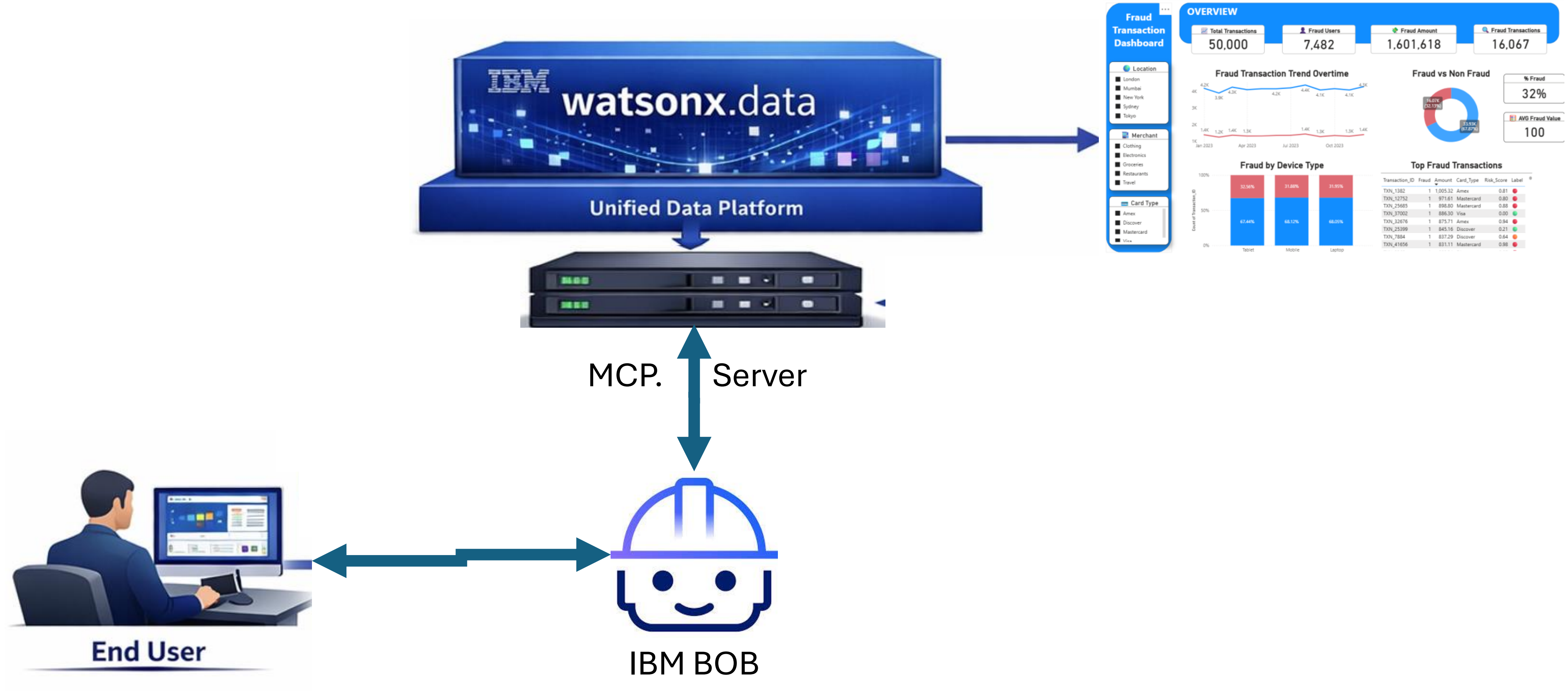
Outcome

- **Fraud losses reduced by 40%** in the first year.
- **Customer trust improved** thanks to fewer false alarms and faster resolution.
- **Operational efficiency increased**, as analysts spent less time chasing false leads and more time on genuine threats.

Future Outlook

- Scaling to **mobile banking fraud**
- Integration with **dark web monitoring**
- Continuous AI model evolution

Solution Architecture



Demo